Time and Gravity Research
GR and QFT connection

The Arrow of Time in a Computable Entropy-Spacetime Framework: Irreversibility, Time Dilation, Global/Local Time, Proper Time and Entropy Coordinates

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ABSTRACT: This study explores the fundamental relationship between time and entropy through a discrete computational framework. We introduce a novel multiplicative entropy definition that inherently guarantees entropy increase while remaining consistent with classical entropy measures. The proposed model establishes a unified mapping between spacetime and entropy, bridging global cosmological time, local relativistic dynamics, and thermodynamic evolution. This approach yields a practical entropy coordinate system for computer simulations of physical processes.The framework derives the arrow of time from the dynamics of space elementary quanta (SEQ) network, offering new insights into temporal irreversibility. It interprets light-speed invariance as a topological property of SEQ propagation and provides a geometric explanation for relativistic time dilation effects. Furthermore, the model suggests testable predictions regarding chiral symmetry breaking in quantum systems and proposes a connection between dark energy and the ground state of the SEQ network. By quantifying entropy production across scales, this work advances the computational modeling of time-dependent phenomena in both classical and quantum regimes.

Key words: multiplicative entropy; space-time-entropy mapping; thermodynamic time arrow; entropy coordinate;

I. INTRODUCTION.

The irreversible nature of time emerges from absolute entropy increase, as demonstrated by our space-time-entropy mapping model. This framework links the arrow of time to multiplicative entropy growth through discrete Space Elementary Quantum (SEQ) networks, proving $\Delta S \geq 0$ universally governs all physical processes. It unifies relativistic time dilation with thermodynamic evolution while explaining phenomena like light-speed invariance and parity violation via fixed SEQ spin chirality

II. METHODS

1 Basic set:

1.1 The universe is expanding.

1.2 The universe operates under the Law of Energy Conservation and the Law of Entropy Increase.

1.3 The universe consists of Space Elementary Quanta (SEQ), forming a topologically homeomorphic 3D structure (adjacency relations remain preserved while individual SEQ energy states may vary). The distortion of light around black holes demonstrates that gravitational and electromagnetic field quanta are coupled, suggesting they originate from the same quantum field in different excited states—leading to the SEQ hypothesis. The spacing and tension between adjacent SEQ can be modulated by gravitational or equivalent gravitational fields.

1.4 All field quanta and elementary particles represent different energy excitation states of SEQ, expressible as 3D dynamic structural matrices of SEQ .

1.5 SEQ possess a ground state energy (e.g., ground-state spin or vibrational modes). If ground-state spin chirality is fixed, this may explain parity violation. The ground-state energy of SEQ could also account for the cosmological constant in General Relativity.

This framework shows strong alignment with Loop Quantum Gravity theory.[1][2]

1.6 Adjacent SEQ maintain a dynamic equilibrium spacing. Perturbations from this equilibrium modify the SEQ's ground resonant frequency and induce associated compressive/tensile forces(origin of gravity and equivalent gravitation). The underlying response function is inherently nonlinear and asymmetric, potentially exhibiting complex behaviors including regional gradient variation. This elastodynamic formulation of spacetime exhibits remarkable conceptual synergy with Einstein's original elastic continuum analogy [3], while providing a concrete realization of Zee's 'spring network' metaphor [4] through the discrete SEQ framework.

The sub-Planckian regime governs the spatial elasticity of the network (including elastic potential storage and release), while quantized energy transfer occurs exclusively through interactions between SEQs. Within this framework:

- The spin degrees of freedom of SEQs and their elastic bonds remain decoupled, preserving independent dynamical regimes.

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- Under perturbation, the system responds by modifying SEQ resonant frequencies while generating compressive/tensile forces.
- This response is nonlinear and asymmetric, enabling emergent behaviors (e.g., regional gradient variations).
- SEQs are stable, indivisible structures composed of sub-Planckian components. SEQs' spin emerges from collective space transformations at the sub-Planck level. This ensures the spin degrees of freedom do not interfere with elastic deformations in the SEQ network. This architecture naturally protects spin dynamics from elastic disturbances.

At the sub-Planckian scale, the elastic properties of the underlying substrate impose an upper bound on the spacing modulation and tension between adjacent SEQs. This fundamental limit ensures that extreme deformations (e.g., near black hole singularities) cannot disrupt the topological integrity of the SEQ network.

1.7 If matter truly traversed space, it would require modification of spacetime's adjacency relations. Yet black holes—despite their extreme mass—preserve local spacetime topology (as evidenced by smooth light bending). This implies that apparent particle motion must instead represent propagating excitations of spacetime itself, consistent with GR. The speed of light (c) constitutes the maximum excitation propagation rate in space.

2 The course of defining time:

2.1 SEQ serve as the electromagnetic wave conducting medium. Matter with mass and its motion are waves in this medium. In this framework, nothing truly moves through space - light speed c is the maximum conduction speed c, preventing velocity stacking beyond c. All physical phenomena correspond to specific energy state configurations, establishing SEQ as the universal substrate.

2.2 The universe's composition: Energy conservation and quantization imply a finite number N of SEQ, each with M possible energy states,(where each energy state is an integer multiple of Planck's constant h,) allowing up to M^N transformations. These M energy states form an algebraic system incorporating translational, spinning and rotational operations. Energy conservation and entropy increase constraints reduce possible transformations significantly below M^N.

2.3 Time definition:

2.3.1 Let J be the possible universe transformations (J <<M^N).

2.3.2 The Planck time (t_p) interval separates adjacent transformations as the minimal time unit.

2.3.3 Time's arrow follows entropy increase.

2.3.4 Transformations map non-bijectively to entropy values (k distinct values partition J transformations into K classes). Parallel transformations share the same entropy values, but only one can occur. The entropy set maps to possible time values - each moment corresponds to one universe transformation. Non-uniform entropy increase means only a subset of possible time values actually occur.

2.3.5 Each space transformation (state transition of the SEQ network) can be assigned a unique entropy value calculated via the multiplicative energy distribution across this space transformation's matrix.

2.3.6 Finite transformations ensure discrete, limited time.

Notably, such discrete evolutionary models find precursor in causal set theory's exploration of discrete spacetime dynamics [5], though our approach differs fundamentally in its entropic time construction."

3 Definition and analysis formula of entropy

In this definition of entropy, energy state norms of the space elementary quantum are used to calculate specific closed space entropy value or the whole universe's entropy value at a given moment (i.e., during a specific state transition). The entropy value can be defined as the multiplicative product of the energy norms of all SEQs involved in that transition(that moment's space transformation).

Entropy value of closed system S= $\prod_{i=1}^{n} \boldsymbol{m_i}, \boldsymbol{i} \in \boldsymbol{N}$ ①

Energy of closed system=constant=$\sum_{i=1}^{n} \boldsymbol{m_i}, \boldsymbol{i} \in \boldsymbol{N}$ ②

$S_{\max} \leq \boldsymbol{m_i}^{\boldsymbol{n}}$, When all m_i are equal or differ only by Planck's constant h

(Where $\boldsymbol{m_i}$ refers to the energy carried by the ith SEQ during a single transformation of the closed system, where each energy state is an integer multiple of Planck's constant h,)

3.1 Energy transfer rules and triggering conditions:

Energy exchange occurs between adjacent SEQ (i,j) if and only if the following thermodynamic gradient exists: $m_i > m_j + h$, Energy transfer occurs only in discrete quanta of Planck's constant h, $m_i \rightarrow m_i - h$; $m_j \rightarrow m_j + h$ *(Planck's constant:h)*

3.2 Logarithmic Relation:

After logarithmic transformation, lnS aligns with the conventional Boltzmann entropy form, while the multiplicative formulation naturally suits discrete systems.

3.3 Proof of Spontaneous Entropy Increase

Theorem:

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For every possible energy transfer process, the total entropy change satisfies $\Delta S \geq 0$.

Proof Outline: Let the pre-transfer states be m_i=a, m_j=b (a>b+h);

The post-transfer entropy ratio is：

S_{t2} / S_{t1} =(a−h)(b+h)/ab=1+h(a−b−h)/ab > 1 (*Planck's constant h)*

3.4 Numerical Example: System States and Entropy Evolution

Table 1: Simplified Entropy Increase Demonstration

(table1：Simplified Entropy Increase Demonstration)

3.5 Entropy increase manifests through macroscopic interactions level, atomic/molecular level, QED processes, and microscopic particle restructuring (matrix disintegration/reorganization, radiation).

3.6 The entropy formulation$= \prod_1^n m_i; i \in N$ proposed herein operates not only at the fundamental SEQ level, but also admits coarse-graining for arbitrary massive many-body systems. This universality originates from the energy-mass equivalence which allows the cumulative product to naturally encode multi-body interactions. Crucially, this formulation provides first-principles computational constraints for discrete simulations across all scales.

At this point, a clear multiple mapping can be established:

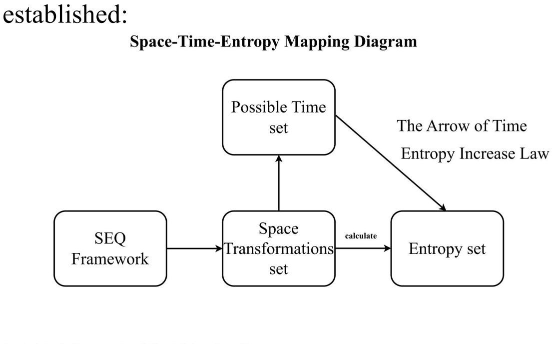

Diagram 1: Space-Time-Entropy Mapping Diagram
Drawing tool: draw.io [10]

3.7 First principle: The spontaneous Entropy Increase is Causal Order.

4 Analysis of Action

The dimension of the action quantity is consistent with that of Planck constant, so the action quantity can be the number of units of quantum energy (Planck constant).

The total number of unit quantum conduction energy of a physical process involves two parameters, one is the number n of SEQ involved in the physical process wave conduction, and the other is the conduction times$k_i$of the ith space elementary quantum involved in the conduction, in which the maximum number of conduction count k_i of all single SEQ is less than or equal to times of local spatial transformations in this process. h is Planck constant. Then the action amount can also be written as

$$\sum_{i=1}^{n} hk_i, i \in N \quad ③$$

The law of the principle of least action reveals that the path selected by physics process is the path that the total amount of energy conduction involved SEQ

System State	SEQ Energy Distribution $\sum_{i=i}^{n} m_i$=12	Entropy $\prod_{i=1}^{n} m$	Remarks
Initial non-equilibrium state	[3, 1, 5, 3]	45	-
Intermediate state	[3, 1, 4, 4]	48	-
Final state	[3, 2, 3, 4]	72	Due to adjacent energy transfer with minimal quanta *h*, this system cannot reach maximum entropy in this case

is the least. It involves two parameters, as mentioned above. If the number of SEQ involved remains the same and the conduction times of each SEQ are the same, (1)Fermat's principle of the shortest time of optical path and (2)the principle of the steepest descent line can be directly derived, because in these two examples, the number of SEQ involved in the physical process and the rolling spherical rigid body in the steepest descent line remain the same, and the conduction times of each SEQ are also approximately the same, then time (the number of local space transformations) is the only variable for the calculation of action, so using time to divide the wave forms of different paths is equivalent to using action to divide the wave forms of different paths. Time here refers to the transformation times of local space. From this point of view, we can probably understand why the analysis of action amount proposed in different periods is different, but it can explain some phenomena.

5 Local time , the proper time and relative time in Relativity

5.1 **local time**. As previously established, global time is defined by the transformations of the universe.

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This work introduces local time as an operational concept that: (1) provides correspondence with special relativistic time notions; and (2) enables precise specification of time scales for localized physical processes. Crucially, any measurable time parameter in physical calculations necessarily corresponds to transformations within a specific local space. This operational concept is designated as **local time**.

The local space scope must be unambiguously specified: either as the SEQ network along the physical process path or the connected region within the observer's light cone. This distinction mirrors the complementarity between Feynman's path-integral formulation and relativistic theory, avoiding conceptual confusion in prior works.

Local time: it can specify the time set corresponding to the transformations within a specific spatial range. In fact, from the perspective of this framework, the existing equations with *t* parameter in textbooks are actually the local time by default.

One example: the clock slowness effect in Relativity manifests as different transformation times in different local spaces.

Every local space transformation constitutes a part of the universal evolution. Global time progression (ΔT_global) does not necessitate synchronous local transformations, the state matrix of a given local space may remain invariant despite cosmic-scale changes ($\Delta\tau\_local \approx 0$) when this local space undergoes no state transition.

5.2 **The proper time in Relativity** is related to local transformations count of physics process entities. (Section 6.5 present the physical mechanism underlying proper time dilation in GR.

5.3 Understanding on **Lorentz-covariant rules** in **Special Relativity** theory: The time perception of physical processes across distinct reference frames fundamentally corresponds to the observation of transformation counts. The observation time discrepancy between frames derives from the accumulated difference in their frame SEQ transformation counts. An observer measures another frame's time evolution through the differential transformation count ΔN, while he can't perceive their own transformation count N_0 . The observed ΔN is fundamentally governed by the dynamic light-path difference between the observer's frame and the moving reference frame of the measured object. Under the principle of non-additivity of light speed (c-invariance), this formulation naturally derives Lorentz-covariant rules through counting operations.

Key Distinction from GR Effects

While the mathematical derivation process aligns with standard special relativity textbooks—replacing continuous spacetime metrics with discrete counting operations—the physical interpretation differs substantially in conceptualization:

- SR Effects as Perceptual Phenomena

Time dilation and length contraction emerge purely as observer-dependent measurement consequences.

Originate from information transmission constraints via discrete light-signal counting.

- Contrast with GR Mechanisms

Gravitational time dilation involves actual deformation of the SEQ network's transformation frequency.

Equivalence principle effects require space compression as well.

6 Phenomenological consistency checks

6.1 **Why can't the speed of light stack up?**

As established above, the speed of light (c) constitutes the maximum excitation propagation rate in space, wherein all observed motion fundamentally represents state transitions within this immutable lattice rather than physical traversal through space.

6.2 **Uncertainty Relation and Wave-particle Duality.**

The Uncertainty Principle naturally arises from wave propagation through discrete SEQ: precise position inherently limits determination of conduction speed (wave dynamics), and vice versa.

Wave-particle Duality. The wave nature is fundamental, while the particle nature emerges from the discreteness of spacetime itself.

6.3 **Double-slit experiment.** [6][7]

Explanation: Based on this framework, electron's conducting in the space conforms some probability function, and really causes multi-path oscillating in space, and these oscillating can be accumulated. Electron excitation causes space oscillating both of the two slits at the sides of excitation source, The proposed framework predicts the emergence of interference fringes, as the electron excitation induces coherent oscillations across multiple SEQ conduction paths. Even in case of one-slit experiment, when the accumulated Wavelets cross the slit, slit's Unsmooth edges may produce different paths and generate interference that could be observed if the sensor is

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sensitive enough and the slit edge's form can meet the Coherent Condition.

Note:While this framework strictly enforces time ordering along entropy increase, it currently does not provide an interpretation for delayed-choice experiments. This remains an open question requiring further development in this framework.

6.4 **Non-conservation of parity**

With the assumption above, if SEQ ground state spin chirality is fixed, that could be one possible explanation of the non conservation of parity.

Is dark matter potentially explained by high-density SEQ ground-state clusters under gravity? And is the ground state energy carried by the SEQ the so-called dark energy?

6.5 **Conjecture on Muon Decay experiment[8]**

An object's motion (velocity/acceleration) deforms the SEQ network via:(1)stress-energy dependent spacing variation,(2) strain-modulated transformation frequencies , and (3) GR-consistent effects (velocity-dependence mirrors SR, acceleration induces curvature). Particle decay results from destabilization of their 3Dstructure matrices when interaction forces (EM/weak/strong) fail to maintain equilibrium.

7 **A prediction of a difference in the magnetic moments of the positron and electron(falsifiable)**

Given the isotropic nature of the electric field generated by electrons, this framework hypothesizes that electrons possess an Spatially Symmetric structure composed of SEQ. Under the SEQ framework, all charged microscopic particles including electrons and quarks possess dynamic three-dimensional structures that preserve spherical symmetry in space.
If a slight difference is measured between the magnetic moment of the positron and that of the electron, it would strengthen the credibility of the hypothesis that the SEQ has a ground-state spin. This difference arises because the positron's structural matrix rotates with opposite chirality to the electron's, resulting in distinct coupling configurations with the SEQ's fixed-chirality ground-state spin. Based on this, it can be inferred that the magnetic moments of the positron and electron should exhibit a slight discrepancy. A precision measurement of the positron magnetic moment, currently underway by the Fan-Myers-Sukra-Gabrielse collaboration , could test two key hypothesis of the SEQ framework: the fixed chirality of SEQ ground-state spin and electron's substructure.[9]

III. RESULTS

1.The study successfully establishes a time-entropy mapping model within the discrete spacetime framework, where each temporal state corresponds to a unique entropy value. This mapping ensures that entropy either increases or remains constant over time, providing a clear thermodynamic arrow of time.
2.However, the mapping is not bijective.
3.The model has undergone consistency checks against most key phenomena, confirming its physical plausibility, while proposing a falsifiable criterion.

IV. DISCUSSION

While the present model does not yet incorporate electromagnetic interactions or deeper algebraic structures, it establishes a foundational framework for such future developments.We envision that further development using an algebra system and discrete differential geometry could refine this framework, ultimately achieving better alignment with both general relativity and the Standard Model.
Within this framework, gravitational interaction emerges as a translational operation that modulates the inter-SEQ spacing and consequently alters the stress-energy distribution; the emergent gravitational field of mass exhibits spherical symmetry due to SU(3) color charge distribution[11]. Serving as the symmetry-preserving mechanism at quantum scale, While this manuscript focuses on space-time-entropy mapping,

V. CONCLUSIONS

This work presents a theoretical framework whose mathematical formulation, while deliberately minimalist, yields tractable derivations and falsifiable predictions. The model demonstrates particular utility in elucidating the fundamental relationship between temporal evolution and entropy, with immediate applications to computational simulations through the embedding constraints of thermodynamic laws and least-action path selection offering an entropy coordinate for computer evolution simulations.

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